Course name: Bayesian Inference
Duration: 40 minutes
Maximum marks: 10

1. Let X has probability density function

$$f(x \mid \beta) = \frac{x \exp(-x/\beta)}{\beta^2}, \qquad x > 0,\ \beta > 0.$$

   Find a one-to-one reparameterization $\phi = h(\beta)$ such that the Jeffreys prior for ϕ is uniform, i.e.,

$$\pi_J(\phi) \propto 1.$$

   [5]

2. Consider two coins, C1 and C2 , with the following characteristics: $Pr(heads|C1) = 0.7$ and $Pr(heads|C2) = 0.3$. Choose one of the coins at random and imagine spinning it repeatedly. Given that the first two spins from the chosen coin are heads, what is the expectation of the number of additional spins until a tail shows up? [5]

Solution

1. We have

$$f(x \mid \beta) = \frac{x \exp(-x/\beta)}{\beta^2}, \qquad x > 0,\ \beta > 0.$$

   The log-likelihood for one observation is

$$\ell(\beta) = \log x - \frac{x}{\beta} - 2\log\beta.$$

   Therefore,

$$\frac{\partial \ell}{\partial \beta} = \frac{x}{\beta^2} - \frac{2}{\beta},$$

   and

$$\frac{\partial^2 \ell}{\partial \beta^2} = -\frac{2x}{\beta^3} + \frac{2}{\beta^2}.$$

   Since

$$E_\beta(X) = 2\beta,$$

   the Fisher information is

$$I(\beta) = -E_\beta\left[\frac{\partial^2 \ell}{\partial \beta^2}\right] = \frac{2}{\beta^2}.$$

Hence, the Jeffreys prior for β is

$$\pi_J(\beta) \propto \sqrt{I(\beta)} \propto \frac{1}{\beta}.$$

Let

$$\phi = h(\beta)$$

be a one-to-one reparameterization. Under this transformation,

$$\pi_J(\phi) = \pi_J(\beta) \left| \frac{d\beta}{d\phi} \right|.$$

For $\pi_J(\phi)$ to be uniform, we require

$$\frac{1}{\beta} \left| \frac{d\beta}{d\phi} \right| \propto 1.$$

Equivalently,

$$\left| \frac{d\phi}{d\beta} \right| \propto \frac{1}{\beta}.$$

Thus,

$$\phi = c \log \beta + d, \qquad c \neq 0.$$

A simple choice is

$$\boxed{\phi = \log \beta}.$$

Indeed, since $\beta = e^{\phi}$,

$$\pi_J(\phi) \propto \frac{1}{e^{\phi}} \frac{d(e^{\phi})}{d\phi} = 1.$$

Hence, the Jeffreys prior for ϕ is uniform.

2. Let A denote the event that the first two spins are heads. Initially,

$$P(C_1) = P(C_2) = \frac{1}{2}.$$

Since

$$P(A \mid C_1) = (0.7)^2 = 0.49$$

and

$$P(A \mid C_2) = (0.3)^2 = 0.09,$$

Bayes' theorem gives

$$P(C_1 \mid A) = \frac{(1/2)(0.49)}{(1/2)(0.49) + (1/2)(0.09)} = \frac{49}{58},$$

and hence

$$P(C_2 \mid A) = \frac{9}{58}.$$

Let N denote the number of additional spins until the first tail occurs. Given that C_1 was selected, the probability of a tail on each subsequent spin is 0.3. Therefore,

$$E(N \mid C_1, A) = \frac{1}{0.3} = \frac{10}{3}.$$

Similarly, given that C_2 was selected, the probability of a tail is 0.7, and hence

$$E(N \mid C_2, A) = \frac{1}{0.7} = \frac{10}{7}.$$

Therefore, by the law of total expectation,

$$\begin{aligned} E(N \mid A) &= P(C_1 \mid A)E(N \mid C_1, A) + P(C_2 \mid A)E(N \mid C_2, A) \\ &= \frac{49}{58}\frac{10}{3} + \frac{9}{58}\frac{10}{7} \\ &= \frac{535}{203}. \end{aligned}$$

Thus,

$$\boxed{E(N \mid A) = \frac{535}{203} \approx 2.636.}$$